Git Commands » **CHEATSHEET**

People, Data, & Drugs
Center for Pharmaceutical Data Science Education

You can operate Git in the terminal after installing it from **git-scm.com**.
To verify that Git is installed and check its version:
`git --version`

If unsure, use git help:
`git command --help`

Don't like the terminal? Use an app like *GitHub Desktop* instead.

Create

Initialize a new repository locally or clone an existing one.

Initialize new repository in the current directory:
`git init`

Initialize new repository in a specific directory:
`git init <directory>`

Download from an existing repository:
`git clone <url>`

Inspect

List uncommitted new and changed files:
`git status`

Display changes not yet staged:
`git diff`

Branches

Display local branches:
`git branch`

List local and remote branches:
`git branch -av`

Create a new branch:
`git branch <branch_name>`

Delete branch:
`git branch -d <branch_name>`

Switch to branch <branch_name>:
`git switch <branch_name>`

Create and switch to a new branch:
`git switch -c new-branch`

Merge <branch_a> into <branch_b>:
`git switch <branch_b>`
`git merge <branch_a>`

Changes

Stage a file:
`git add <file_name>`

Stage all changed files:
`git add .`

Commit all staged files:
`git commit -m "commit message"`

Unstage file, keeping file changes:
`git reset <file_name>`

Undo uncommitted changes:
`git restore <file_name>`

Undo a commit:
`git revert <commit>`

Synchronize

Get the latest changes from the remote repository (no merge):
`git fetch`

Fetch the latest changes from the remote repository and merge:
`git pull`

Upload local changes to the remote repository:
`git push`